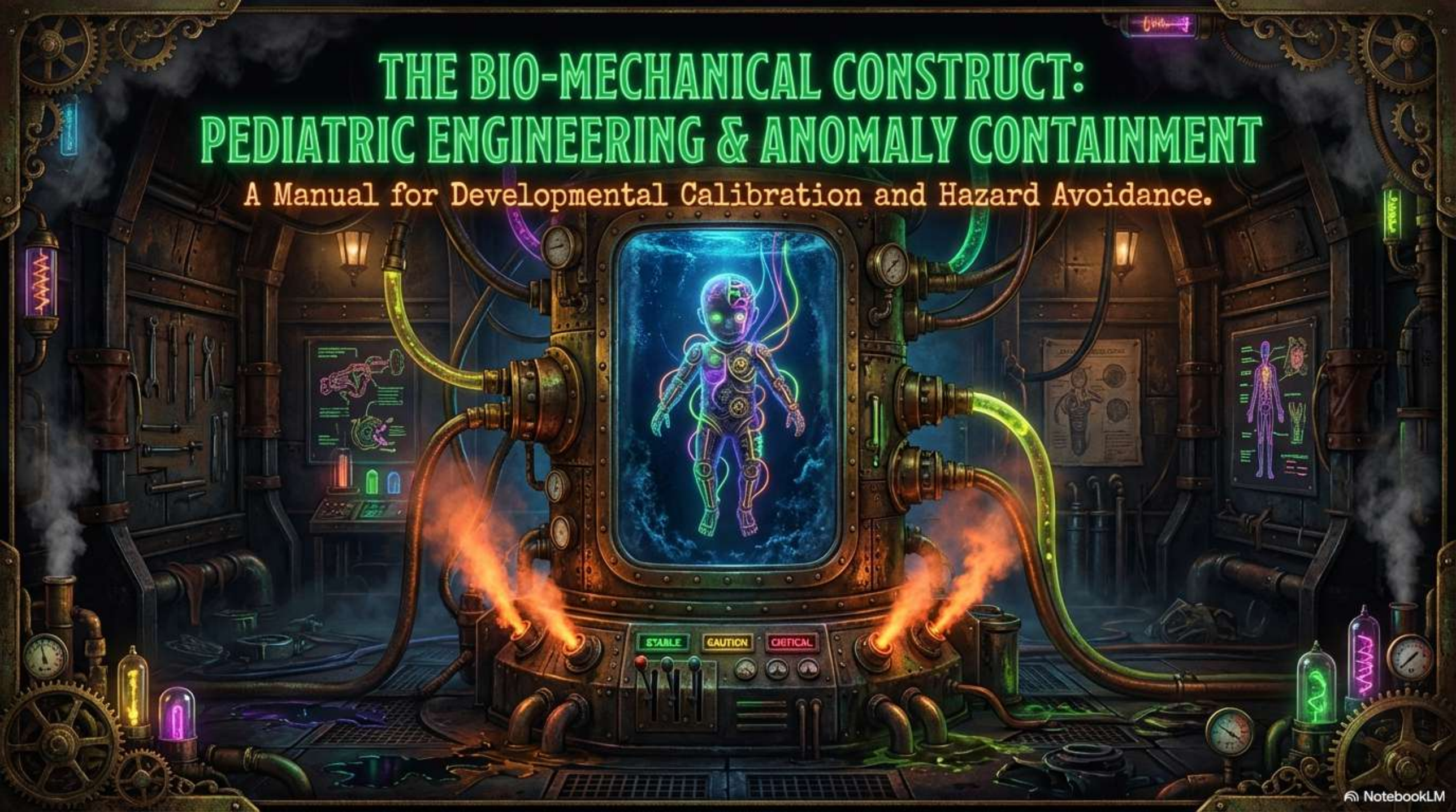
THE BIO-MECHANICAL CONSTRUCT: PEDIATRIC ENGINEERING & ANOMALY CONTAINMENT

A Manual for Developmental Calibration and Hazard Avoidance.



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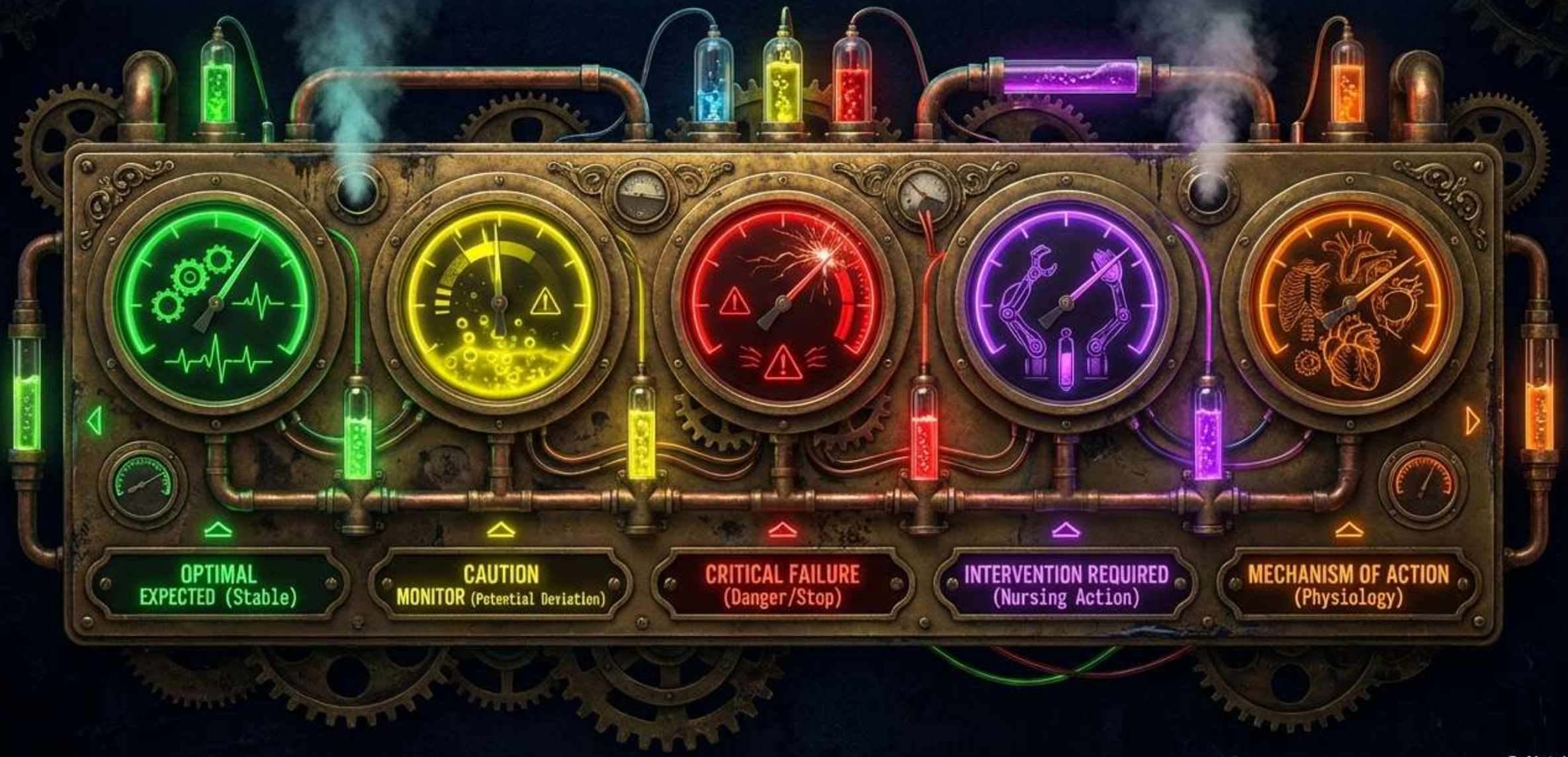
OPERATOR'S LEGEND: SYSTEM STATUS INDICATORS



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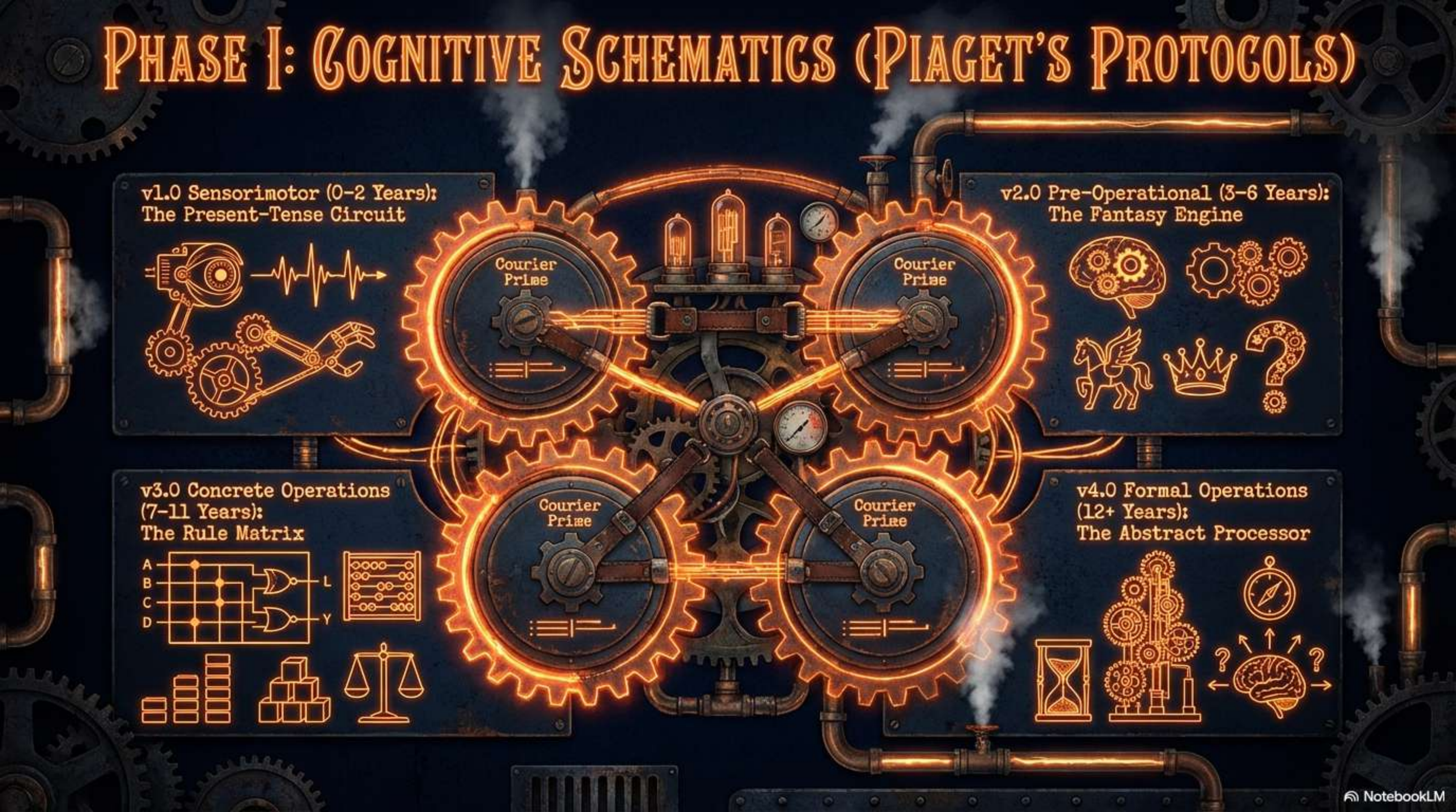
Phase I: Cognitive Schematics (Piaget's Protocols)



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V1.0 Sensorimotor (0-2 years): Real-time Processing

- LOGIC: Construct only understands the NOW. No Past/Future data.
- ACTION: Teach WHILE you perform the procedure.
- WARNING: DO NOT use pre-operative preparation.
- Verbal Protocol: “I am lifting your arm now.” (Present Tense Only)

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v2.0 PRE-OPERATIONAL (3-6 YEARS): THE FANTASY ENGINE

LOGIC: Imaginative, Illogical, Magical Thinking.

ACTION: Teach the MORNING OF the procedure.

MECHANISM: Learn through PLAY (Puppets, Dolls).

CAUTION: Avoid delays. High risk of imagining the worst.

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v3.0
CONCRETE OPERATIONS (7-11 YEARS): THE RULE MATRIX

LOGIC: Rule-Oriented. Rigid. Only one way to do things.

ACTION: Teach DAYS AHEAD.

EXPECTED: Use Audio/Visual materials & age-appropriate reading.

MNEMONIC: “7-11 Grocery Stores are surrounded by concrete.”

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v4.0 FORMAL OPERATIONS (12+ YEARS): THE ABSTRACT PROCESSOR

- LOGIC: Abstract thinking. Cause and Effect analysis.
- ACTION: Teach like an ADULT.
- EXPECTED: Can MANAGE their own care.

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THE THRESHOLD OF AUTONOMY: DEFINING 'MANAGEMENT'

DEFINITION: 'Managing' = Decision making based on abstract analysis.

10-Year-Old CANNOT manage a scrape (Wash/Dry/Bandage is a RULE, not a decision).

12-Year-Old CAN manage Chronic Renal Failure (Will detect missed bruit and seek help).

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Phase II: Calibration Tools (Toy Safety Protocols)

1. SAFE: No small parts < 4 years (Aspiration Risk).
2. FEASIBLE: Can the construct perform the task? (e.g., No swimming in body cast).
3. NO METAL TOYS if Oxygen is in use.
4. FOMITES: Stuffed animals = Bio-Magnets.
5. HARD PLASTIC: Best for immunosuppressed (Disinfectable).

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Calibration: 0-6 months (Sensorimotor)

BEST CHOICE: Musical Mobile.

MECHANISM: Stimulates Motor & Sensory input.

ALTERNATIVE: Soft and Large objects.

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Calibration: 6-9 months (Object Permanence)

MECHANISM: Teaching object existence when unseen.

BEST CHOICE: Cover/Uncover Toys (Jack-in-the-Box, Peek-a-Boo).

DANGER: NO MOBILES after 6 months (Strangulation Risk).

TEXTURE: Large and Firm.

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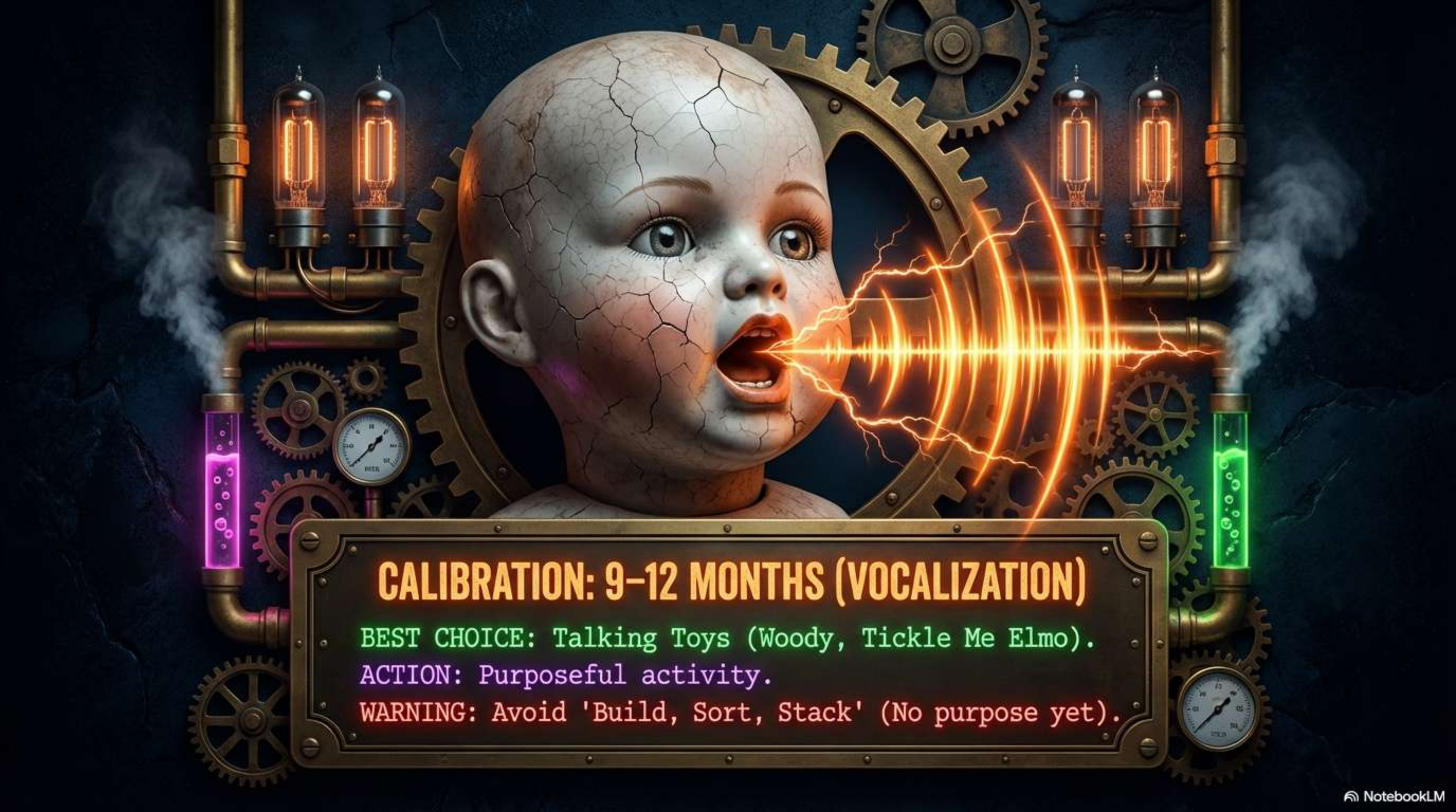
CALIBRATION: 9–12 MONTHS (VOCALIZATION)
BEST CHOICE: Talking Toys (Woody, Tickle Me Elmo).
ACTION: Purposeful activity.
WARNING: Avoid 'Build, Sort, Stack' (No purpose yet).

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Calibration: Toddlers 1-3 Years (Gross Motor Hydraulics)

BEST CHOICE: Push/Pull Toys (Wagons, Lawnmowers).

MECHANISM: Gross Motor (Running, Jumping).

CAUTION: NO finger dexterity yet (No pencils/scissors).

PLAY STYLE: Parallel Play (Alongside, not with).

EXCEPTION: Finger Painting is “Hand Painting”.

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CALIBRATION: PRESCHOOL 3-6 YEARS (FINE MOTOR & BALANCE)

BEST CHOICE: Tricycles, Dance, Skates (Balance).

MECHANISM: Fine Motor Dexterity.

PLAY STYLE: Cooperative Play (Group operations).

KEY FEATURE: Highly Imaginative / Pretend.

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Calibration: School Age 7-11 Years (The 3 C's)

Courier Prime

1. Creative (Make it). 2. Collective (Gather it).
3. Competitive (Win it).

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CALIBRATION: ADOLESCENTS (PEER NETWORKING)

Courier Prime

EXPECTED: Peer Group Association.

ACTION: Allow peers in room.

THE EXCLUSION PROTOCOL (Bar Visitors If):

1. Fresh Post-Op (<12 hours)
2. Immunosuppressed
3. Contagious Disease

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SYSTEM ANOMALY: CLEFT LIP/PALATE REPAIR

CRITICAL: NOTHING IN THE MOUTH.

PROHIBITED: Suction, Tongue Depressors, Thermometers, Straws, Spoons.

REASON: Disruption of suture line (Intake Valve Failure).

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System Anomaly: Epiglottitis (Airway Occlusion)

CRITICAL: NEVER use a tongue blade to evaluate.

MECHANISM: Irritation causes complete spasm/shut-down.

SYMPTOMS: Drooling, dysphagia, stridor.

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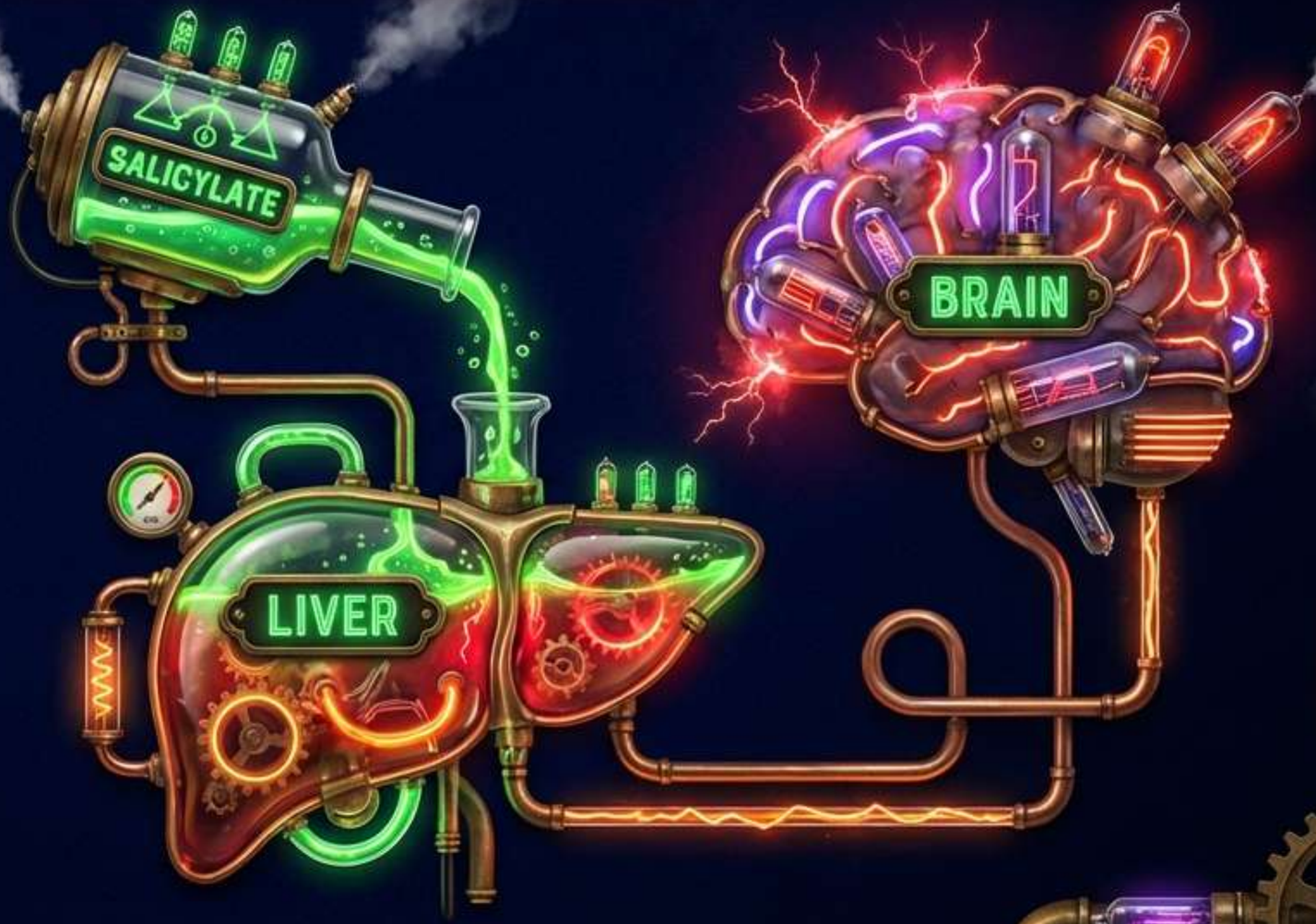
SYSTEM TOXICITY: REYE'S SYNDROME

CAUSE: Aspirin (Salicylates) given to children with viral infections.

RESULT: Hepatic Encephalopathy.

MECHANISM: Liver dysfunction + Brain swelling.



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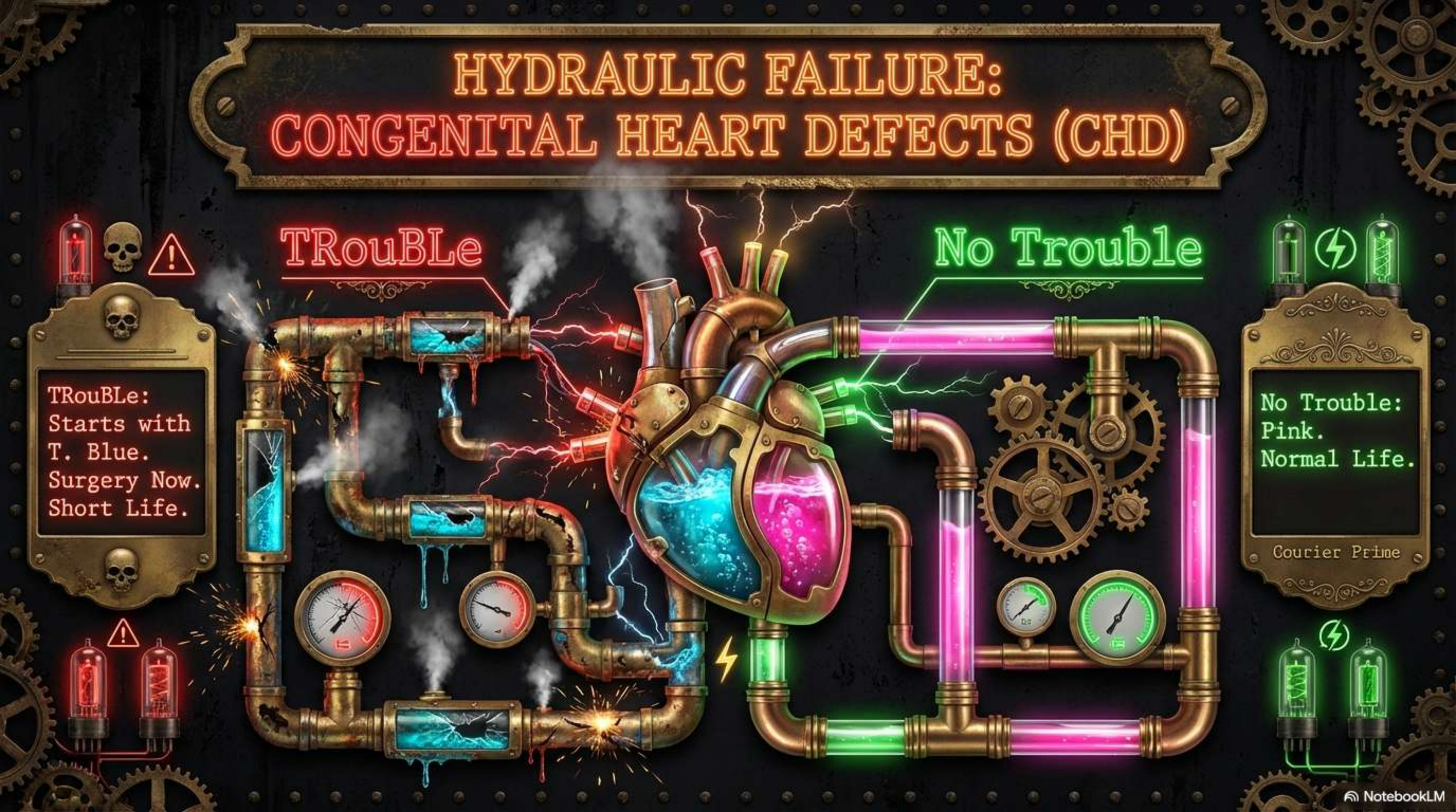
HYDRAULIC FAILURE: CONGENITAL HEART DEFECTS (CHD)



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The Defect Cluster: Tetralogy of Fallot

VarieD PictureS Of A RancH

V-D: **VENTRICULAR DEFECT:** Hole in the wall between ventricles.

P-S: **PULMONARY STENOSIS:** Narrowing of the pulmonary valve/artery.

O-A: **OVERRIDING AORTA:** Aorta is shifted, opens into both ventricles.

R-H: **RIGHT HYPERTROPHY:** Thickened right ventricular wall.

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